



MOIZ DAWOODI
O/A LEVEL PHYSICS

O' LEVEL PHYSICS

Mass & Weight

1.3

STUDENT NAME



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PHYSICS WITH
MOIZ DAWOODI

1.3 Mass and weight

- 1 State that mass is a measure of the quantity of matter in an object at rest relative to the observer
- 2 State that the mass of an object resists change from its state of rest or motion (inertia)
- 3 Know that weights, and therefore masses, may be compared using a beam balance or equal-arm balance
- 4 Describe how to determine mass using an electronic balance
- 5 Describe how to measure weight using a force meter
- 6 Define gravitational field strength as force per unit mass; recall and use the equation

$$\text{gravitational field strength} = \frac{\text{weight}}{\text{mass}}$$

$$g = \frac{W}{m}$$

and know that this is equivalent to the acceleration of free fall

- 7 State that a gravitational field is a region in which a mass experiences a force due to gravitational attraction

Mass:

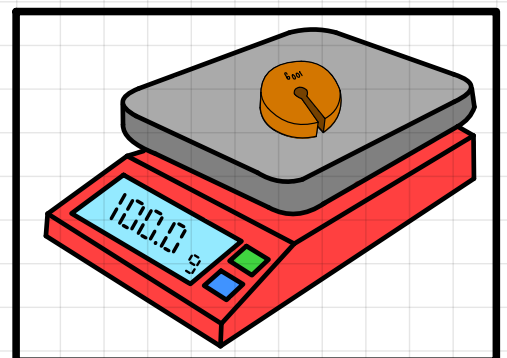
"Amount of matter present in a body relative to observer is called mass."

The S.I unit of mass is kilogram (kg).

Mass remains constant throughout the universe.

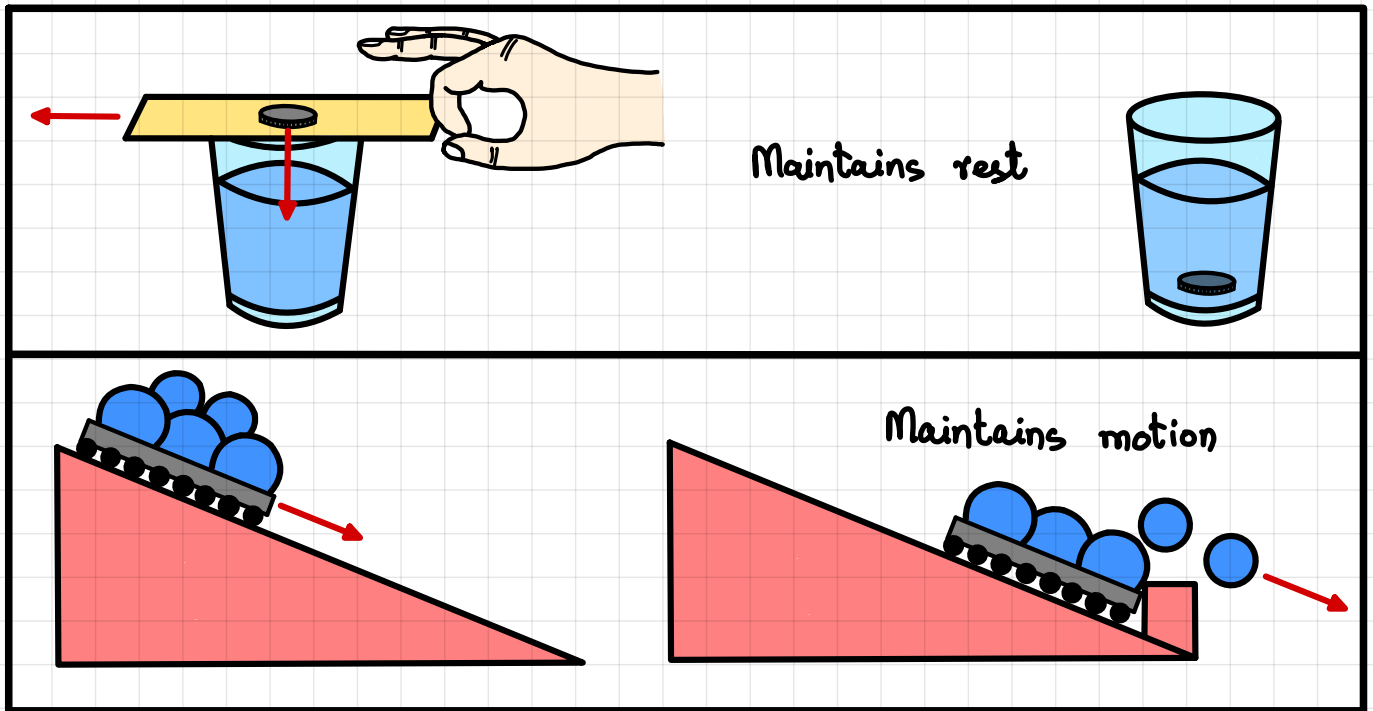
Its a scalar quantity.

It is measured by electronic balance.



INERTIA:

"Property of mass which resist the change in its state."



WEIGHT:

"A force which pulls the body downwards towards Earth."

Its S.I unit is Newton(N).

It changes with gravity.

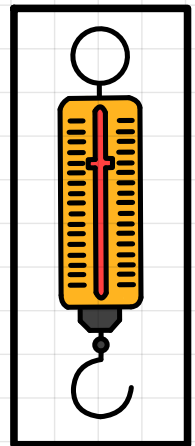
Its a vector quantity.

It is measured by spring balance.

FORMULA:

Weight = Mass $\times$ Gravitational Field Strength

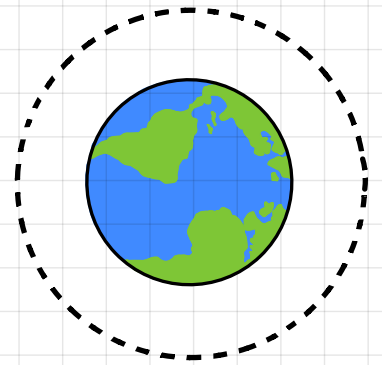
$$W = m g$$



GRAVITATIONAL FIELD STRENGTH:

The space around the Earth in which Earth applies gravitational pull on other objects is called gravitational field.

"Gravitational field strength is the force per unit mass."



FORMULA:

$$\text{Gravitational Field Strength} = \frac{\text{Force}}{\text{Mass}}$$

$$g = \frac{F}{m}$$

- The gravitational field strength of Earth is 9.8 N/kg & the acceleration of free fall is 9.8 m/s^2 .

MASS VS WEIGHT:

MASS	WEIGHT
• Quantity of matter present in a body. • S.I unit = kilogram (kg). • Measured by electronic balance. • Scalar quantity. • It remains constant throughout universe.	• The force which pulls the body downward towards Earth. • S.I unit = Newton (N). • Measured by spring balance. • Vector quantity. • It changes with gravity of planets.



MATCH THE FOLLOWING



Mass is the

throughout the universe.

Mass remains same

amount of matter in a body.

Weight is the force

gravity.

Weight changes with

earth applies the body towards the center.

Inertia is the property of

mass which resist the change in state.